

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Equipment and Material Preservation Procedure

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Revision History

Rev No.	Date	Description of Revision
R01	05.07.22	Issued for Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AG	Associated Gas
ANSI	American National Standards Institute
API	American Petroleum Institute
BBOU	Bubouwe Bou Manifold
BFPC	Brass Fertilizer & Petrochemical Company Limited
BLV	Blowdown Valve
CAO	Computer Aided Operations
ESD	Emergency Shut Down
FEED	Front End Engineering Design
FLB	Field Logistics Base
FLM	First Line Maintenance
FQI	Flow Quantity Indicator
GSPA	Gas Supply & Purchase Agreement
HP	High Pressure
HSSE	Health Safety Security and Environment
LCP	Local Control Panel
LCV	Level Control Valve
LP	Low Pressure
MSDS	Material Safety Data Sheet
MTPA	Metric Tonnes per Annum
MTPD	Metric Tonnes Per Day
NAG	Non-Associated Gas
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Operating Mining License

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opubene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

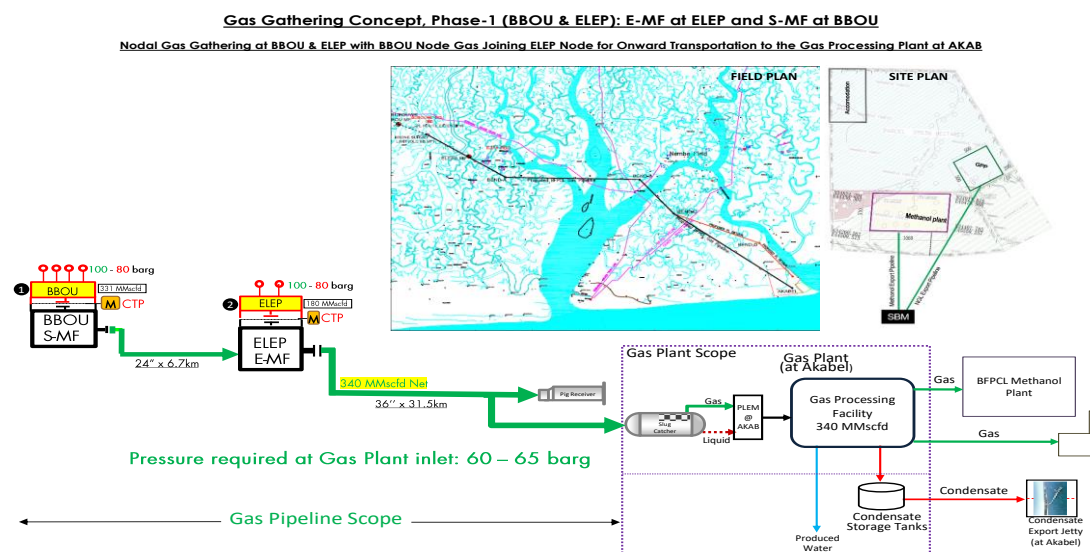


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to define the preservation methods that would be adopted for materials and equipment on the BFPCL gas gathering pipelines and manifolds project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. RESPONSIBILITIES

3.1. Project Procurement Manager

The Project Procurement Manager has a responsibility for the procurement of all materials, shipment to Nigeria and delivery to the project materials storage site in good condition. He is also responsible for transferring all the procurement information and documentation to the site Material Management Team for the effective control of material.

3.2. Project Materials Manager

The Materials Manager has the responsibility for ensuring the implementation of the Equipment and Material Preservation Procedure, and control of equipment and materials from receiving to issue for Fabrication/Construction.

3.3. Construction Base Manager

The Construction Base Manager has a responsibility for ensuring all the site material control work shall be planned, prepared and executed effectively not to affect any other construction activity. The Construction Base Manager shall monitor all the material control activity from the arrival to the issue and review and advise to site material control team. Also has a responsibility for the expediting of disposition of damaged and excepted material not to impact construction schedule.

The Construction Base Manager is responsible to prepare the special handling, storage and preservation procedures, when required.

3.4. Quality Control Manager

The QC Manager has a responsibility of the quality control activity work of the site material control work. During the receiving, storage, issue and return of material, the quality control manager shall execute quality control activity separately and cooperatively with material control activity executed by material control team.

4. RECEIPT, STORAGE HANDLING AND ISSUE

4.1. Equipment and Material Receipt

In addition to shipping notices, the storekeeper should receive packing lists before equipment and materials arrive at site in order to plan for unloading and to ensure adequate storage areas are prepared.

As soon as the material arrives, the storekeeper shall make a first check to ensure that the package corresponds with the packing list and that the material (as far as can be inferred from their outward appearance) are in good condition. Any discrepancy shall be reported on the appropriate report form.

Additional inspection shall be performed by the QC Engineer in accordance with the applicable Receipt Inspection Procedure. All items are on HOLD until QC inspection has been performed.

All receiving inspection activities shall be recorded on the Material Receipt Report (MRR) and Daily Receipt Summary Report shall be issued to advise material received on a daily basis and shall be distributed to Fabrication, Construction and Procurement teams, in addition to the client BFPCL.

Any material delivered with damage or discrepancy shall be handled in accordance with the procedures for the control of Non-Conforming Item.

The Material Management team shall establish, at the construction site, a material handling and storage area with fencing, office/ storage facility and handling facility for the purpose of materials protection, preservation and control.

Upon receipt of materials at the site, the Material Management Team shall prepare an inspection and verification report of the condition and contents listing of the materials. Inspection shall be undertaken by qualified personnel and shall include verification of dimensional correctness and material certification, and checking of shortage and damage.

All the receipt inspection activities shall be recorded and maintained in accordance with the Receipt Inspection Procedure.

When equipment or material is received defective or otherwise unsuitable because of vendor error or in-transit damages, or in those cases where subsequent activities (installation, start-up, etc.) reveal deficiencies, the vendor or freight carrier, as required, will be notified as soon as possible. Every effort will be made to give the responsible party the opportunity to correct the deficiency or provide detailed report before Site Material Manager and/or Project Procurement Manager proceeds with the necessary corrective action.

4.2. Equipment and Material Storage

Upon receipt of the MRR approved by the QC Manager, the Material Controller shall register the

accepted items on the Material Inventory Check List and store them in one of the facilities described below. The storage level for each item shall be defined prior to storage in accordance with vendor recommendation.

Material Controller shall establish material and equipment storage plan and areas, under its direct control, at all Site locations taking into consideration the various storage requirements. The storage facilities are classified as follows:

- Level A: Warehouse to provide temperature and humidity control within specified limits
- Level B: Warehouse to provide temperature control within specified limits
- Level C: Warehouse not to provide temperature and humidity control
- Level D: Yard storage

Storage methods and procedures shall comply with the following requirements:

- All materials shall be stored in such a manner as to permit ready access for inspection or maintenance without excessive handling to minimize risk of damage.
- Materials stacked for storage shall be arranged so that racks, cribbing, or crates are bearing the full weight without distortion of the material.
- Hazardous chemicals, paints, solvents, and other materials of a like nature shall be stored in well-ventilated areas.
- Materials and their containers shall be plainly marked so that they are easily identified without excessive handling or unnecessary opening of crates and boxes.
- Weatherproof coverings, when used for outdoor storage, shall be the flame-resistant type of sheeting or tarpaulins. They shall be placed so as to provide drainage and to ensure air circulation to minimize condensation. They shall be tied down to prevent moisture from entering laps and to protect the coverings from wind damage.
- Materials stored outdoors shall be positioned or covered to avoid trapping moisture in pockets or internally. For example, valves shall be positioned such that water does not collect under the bonnet but can drain from the valve packing area.

Storage areas shall be maintained in accordance with applicable requirements and in accordance with the Material Control Procedure and Vendor recommendation.

- Access to storage areas shall be controlled and limited only to personnel designated by the Material Controller and all areas shall be secure against unofficial access.
- The Material Controller shall maintain the Temperature/ Humidity Check Record for the areas which require control of temperature/ humidity.
- Cleanliness and good housekeeping practices shall be enforced at all times in the storage areas. The storage areas shall be cleaned as required to avoid the accumulation of trash, discarded packing materials, and other detrimental soil.
- Fire protection commensurate with the type of storage area and the material involved shall

be provided and maintained.

- The use or storage of food, drinks, and salt tablet dispensers in controlled storage areas shall not be permitted.
- Measures shall be taken to prevent the entrance of rodents and other animals into indoor storage areas or equipment to minimize possible contamination and mechanical damage to stored material.

The Material Management Team shall make a plan that ensures materials for incorporation in the works are segregated from any other materials for material identity and traceability, and that access is possible only via the Material Management team and that all areas are secure against unofficial access.

Hazardous chemicals, paints, solvents, and other materials of a like nature shall be stored in well-ventilated areas and taken special care.

4.3. Equipment and Material Handling

Equipment and materials shall be handled carefully in order to protect them from loss, pollution and deterioration.

The special materials which are subject to damage or deterioration shall be handled in accordance with the approved procedure.

When materials are handled, painted, galvanized or other outside surfaces shall not be damaged.

All machined surfaces of equipment and materials shall be handled carefully in order to prevent damage.

Materials damaged during handling shall be treated in accordance with the procedure for control of nonconforming items to be developed during detailed engineering design.

4.4. Equipment and Material issue and Return

The Construction Engineer shall request the Material Controller to issue materials.

The Material Controller shall issue the materials to the Construction Engineer and record it.

For return of surplus materials, the Construction Engineer shall request inspection to the QC Engineer. The QC Engineer shall review the request for correctness and ensure material identification is legibly marked on the material and check the condition of the material.

The Construction Engineer shall return the reusable left-over items to the Material Controller who will control the returned materials in accordance with the applicable procedures.

Records pertinent to storage shall be maintained by the Material Controller.

Records pertinent to maintenance of the item shall be maintained by the Construction Engineer.

5. PRESERVATION

5.1. General

All equipment and materials on this project shall be preserved to prevent deterioration and ensure their availability at the time of operation.

To enhance this objective, all equipment received shall be stored in a suitable environment that would prevent damage or disintegration, in line with the OEMs recommendation for storage and preservation.

Line pipes shall be preserved against corrosion in line with the appropriate corrosion protection standards. Pipes shall be stored on pipe racks and the open ends capped to prevent reptiles from crawling into the pipes.

Valves, Gaskets, Pressure gauges, Temperature gauges, etc shall be stored in such a condition that would not affect their reliability when they are required.

Electronic components shall be stored in air-conditioned environment to avoid deterioration.

Valves shall be greased and kept in dust free environment.

5.2. Care for Materials and Equipment

Required maintenance during storage shall be implemented in accordance with approved procedures.

Materials in storage shall have all covers, caps, plugs, or other closures intact. Covers removed for internal access shall be immediately replaced and resealed after completion of the purpose for removal.

Items pressurized with inert gas shall be monitored at specified intervals to ensure that the gas pressure is maintained within specified limits during storage. Desiccant humidity indicators shall also be monitored, and desiccants shall be changed or reprocessed as required.

Instrument materials which need energizing shall be energized according to the vendor instruction.

Space heaters enclosed in electrical materials shall be energized.

Rotating electrical equipment shall be given insulation resistant tests on a scheduled basis.

The shafts of rotating equipment shall be rotated on a periodic basis. The degree of turn shall be established so that the parts receive a coating of lubrication, where applicable, and so that the shaft does not come to rest in a previous position (90 deg. and 450 deg. rotations are examples).

Other maintenance requirements specified by the manufacturer's instructions for the material shall be performed.

The Material Controller shall perform inspections to assure that the integrity of the material and its container is being maintained. Non-conformances detected shall be controlled in accordance with the procedure for Control of Nonconforming Item.

5.3. Preservation Program

The preservation program is divided into three categories:

- Temporary preservation (~ 2 months)
- Short-term preservation (< 1 year)
- Long-term preservation (> 1 year)

It is recognized that in some cases there will be justification for adopting a preservation procedure lying between the terms defined above. This will be a matter for local judgement.

5.3.1. Pressure Vessels

Temporary preservation

If vessels are not shipped with inert gas purge, the vessels should be opened, and its internals visually inspected for ingress of water and condition of preservative. Preservative shall be renewed if necessary and flange covers reinstalled. Vessels shall be thoroughly dried, required preservation applied and then sealed using metal flange covers and rubber gaskets on all flanges.

Temporary preservation should be done by using desiccants, preservative powder, or by spraying the internal surface with corrosion preventive fluids.

Oil, grease and organic materials shall not be used for pressure vessels in gaseous oxygen service.

If vessels are shipped with inert purge, then pressure shall be checked and maintained per preservation procedure. Visible weep holes in reinforcing pads, corrosion plates, etc., shall be packed with grease. Machined surfaces, like flange faces other than those furnished with permanent blinds, and other finished or delicate parts shall be well-greased and protected against rusting and damage.

Bolts and nuts shall be coated with a suitable thread lubricant to prevent galling or corrosion. The lubricant shall be removable with mineral spirits or a solvent. Copper or molybdenum disulfide containing thread lubricants shall not be used.

Stainless steel equipment (SS exterior surface) shall be stored separately from CS equipment to prevent getting into contact with CS, CS scales or rust.

Service gaskets and spare parts shall be packaged separately and stored in warehouse.

All packages shall be separately identified by package number with the vessel tag number clearly indicated.

Short/long term preservation

The same actions as for temporary preservation are also required for short- and long-term preservation except using descants or oil.

For short and/or long-term preservation, vessels shall be purged and filled with a dry inert gas such as nitrogen at a pressure of not less than 35kPa (5psi). Pressure gage should be installed at the pressure vessel in order to check the pressure during idle period.

Air reservoirs, pulsation dampeners etc. shall be protected from internal corrosion damage by suspending bags of vapor phase inhibitor in the vessel before it is sealed. Warning labels shall be attached externally, and the quantity of inhibitor bags noted.

For vessel with an internal corrosion resistant cladding (e.g. SS, Nickel-alloys, plastic and rubber or glass-lined equipment) no further measures are required other than as for temporary preservation.

5.3.2. Piping and Fittings

Temporary preservation

Installed piping should be cleaned, dried and sealed.

Fittings and flanges shall be stored in shelter / warehouse so as to minimize ingress of rainwater and other foreign matter.

New piping and fittings (as delivered from vendor) shall be stored appropriately to prevent corrosion.

If these items are in protective coverings (as delivered from vendor) then no further preservation is required.

Short/long term preservation

For installed piping constructed of material other than carbon steel or clad with corrosion resistant material no further measures are required than for temporary preservation.

Carbon steel lines larger than DN 50 (2 in) should be filled with dry air, and then sealed up. Ensure no solid particles or traces of oil can get into the air or inert gas. Where considered justified, the smaller carbon steel lines may be similarly preserved.

Underground lines shall be purged and filled with dry inert gas such as nitrogen at a pressure of not less than 35kPa (5psi) at the shell and tube side. Pressure guages should be installed in order to check the pressure during idle period. Metallic underground piping shall be protected by cathodic protection.

Fittings and flanges shall be spray coated over the entire body using a suitable corrosion resistant fluid. Fittings and flanges of less than DN 200 (8 in) shall be stored in shelter / warehouse so as to minimize ingress of rainwater and other foreign matter. Large fittings may be stored outside for up to 6 months in their delivery crate or a suitable crate supplied at the storage site. The crates

shall be lined with bituminous heavy gauge brown paper. The crates shall be clearly marked with their contents and be raised at least 0.5m (19 in) clear of the ground on timber or other suitable material to avoid immersion in mud and water. A sound tarpaulin or similar waterproof covering shall be placed over the crates and be securely fastened down. Some ventilation shall be permitted at the base and between the crates.

The contents of the crates shall be inspected periodically to check the protective coverings and for any ingress of water.

NOTE: Pipes stored outside warehouse or shelter (as delivered from vendor) shall be stored on a slope to promote drainage. Stored pipe and fittings to be aligned so that the prevailing winds will blow through the equipment to dry it; the higher end of the pipe is at the upstream side except if the prevailing wind is from the south. In that case the higher end is to the north.

5.3.3. Valves and Attachments

Temporary preservation

Valve spindles should be protected with suitable anti-corrosion compound.

Short/long-term preservation

Valves and attachments are preserved together with and in the same manner as the system in which they are incorporated, or by the same procedure applying to the process equipment with which they are associated.

Valves should be left slightly open to avoid metal to metal contact, wherever possible.

Diaphragm valves shall be left in the fully open position.

Valve packing is left in place without further attention during the period of storage.

Valve spindles should be protected with suitable anti-corrosion compound. Only non-acid and no moisture containing lubricants shall be used for the prevention of fretting corrosion.

Suitable preservative powder should be blown into both the pressure and exhaust side of pneumatic diaphragm or piston valve actuators. Openings and main connections of flange shall be sealed vapour tight.

5.3.4. Stuffing Box Packings

Temporary preservation

Leave all packings intact.

Short/long-term preservation

Remove all metal-braided and graphite fibre packings and replace with non-metallic packings. Otherwise, leave intact. Attach a metal tag to the stuffing box to indicate that the packing needs renewal at the time of recommissioning.

5.3.5. Gaskets

New gaskets, bolts and nuts shall be stored in sheltered locations e.g. warehouse (indoors) above dew-point temperature. If these items are in proper original wrapped up condition (as delivered from the vendor) no further preservation is required. Only large or special bolts and nuts (which are very expensive or have long delivery time) should be coated with suitable anti-corrosion oil.

Temporary/short-term preservation

Leave all gaskets intact.

Long-term preservation

Leave all gaskets intact.

5.3.6. Flanges

Temporary preservation

No action required.

Short/long- term preservation

Exposed flanges are vulnerable to corrosion if left unprotected. Clean thoroughly, and then wrap moisture-proof tape round the aperture between adjacent faces of the flange joint. Apply suitable anti-corrosion fluid to nuts and exposed threads of studs.

5.3.7. General Structural Steel Goods

Steel Plates shall be stacked vertically with plastic or treated wood spacers to avoid damage to protective coatings. The bottom rows of tubular, I-beams etc. shall be laid on wooden lined steel racks with a minimum ground clearance of 0.5m (19 in) to avoid contact with water, mud, sand, etc. in order to prevent corrosion. Separate each item and each row by using preservative treated rope/treated wood spacers to prevent metal-to-metal contact, the buildup of water/debris and also to allow a passage of air for circulation.

Where possible, materials shall be stored under shelter. Goods shall be arranged and stored according to their size, grade, type with the preservation date and heat number clearly stencilled in a visible location.

Rotation of stock materials shall follow a "First-in First-out" system.

5.3.8. Instrumentation Equipment

Instrumentation equipment includes instrument units and components, instrument systems, control valves, computer equipment and peripherals, plant communication equipment and systems.

A differentiation is made between instrumentation locally installed in the plant (including local cabinets), further referred to as 'plant instrumentation', and instrumentation installed in control

rooms, basements, auxiliary rooms and analyser houses, further referred to as 'indoor instrumentation'.

Instrumentation equipment will be stored in strict accordance with the manufacturer's specifications.

5.3.9. Electrical Equipment

Electric equipment shall be stored in a properly conditioned warehouse.

Receptacles and panels shall be kept dry by using drying agents, such as silica gel, in order to prevent moisture damage. The desiccant should be placed in a suitable container and periodically renewed.

In all cases, reference should be made to the manufacturer's instructions for specific preservation measures.

5.3.10. Equipment Packages (e.g. Chemical Injection Package)

Equipment packages shall be stored in properly conditioned storage areas and preserved in full compliance with the manufacturer's instructions.